



Understanding Digital Image Processing

VIPIN TYAGI



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Preface



Digital Images are an integral part of our digital life. Digital images are used in every aspect of our daily life. Digital image processing concerns with techniques to perform processing on an image, to get an enhanced image or to extract some useful information from it to make some decisions based on it. Digital image processing techniques are growing at a very fast speed. This book, *Understanding Digital Image Processing* aims on providing digital image processing concepts in a simple language with the help of examples. The major objective of this text book is to introduce the subject to the readers and motivate for further research in the area.

To explain the concepts, MATLAB® functions are used throughout the book. MATLAB® Version 9.3 (R2017b), Image Acquisition Toolbox Version 5.3 (R2017b), Image Processing Toolbox, Version 10.1 (R2017b) are used to create the book material. My thanks to MathWorks for providing support in preparation of this book.

The functions provided by Image Processing Toolbox™ are given in [Appendix A](#). The details of these functions are available at on the website <https://in.mathworks.com/help/images/index.html>

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'C' language is a very popular programming language. Various functions written in 'C' language are given in [Appendix B](#). My sincere thanks to Mr. Dwayne Phillips for permitting to use the code written by him. Related sub-functions and their descriptions are available at <http://homepages.inf.ed.ac.uk/rbf/BOOKS/PHILLIPS/>.

A glossary of common image processing terms is provided in [Appendix C](#).

A bibliography of the work in the area of image processing is also given at end of the book. We are thankful to all authors whose work has been used in preparation of the manu-script. We have tried to include all such work in bibliography, but if we have skipped some work by error, we will include in next version of the book, Color version of some figures have been provided at the end of the book. These figures also have corresponding black & white versions in the text.

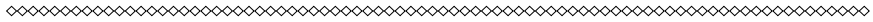
The target audience spans the range from the undergraduate with less exposure to subject to research students interested in learning digital image processing. Many texts are available in the area of digital image processing. In this book, objective is to explain the concepts in a very simple and understandable manner. Hope this book will succeed in its aim.

This work would not have been possible without the help and mentoring from many, in particular, my teacher Prof. Vinod K. Agarwal, Meerut. Special thanks to my dear scholars Mr. K. B. Meena, Dr. Ashwani Kumat and Dr. Deepshikha Tiwari, for their help and support in preparation of the manuscript.

The research work of several researchers contributed to a substantial part of some sections of the book. I thankfully acknowledge their contributions.

It has been a pleasure working with Taylor and Francis Publishers in the development of the book. Thanks to Mr. Vijay Primlani for his kind and timely support in publishing the book and for handling the publication.

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1

Introduction to Digital Image Processing

1.1 Introduction

In today's digital life, digital images are everywhere around us. An image is a visual representation of an object, a person, or a scene. A digital image is a two-dimensional function $f(x, y)$ that is a projection of a 3-dimensional scene into a 2-dimensional projection plane, where x, y represents the location of the picture element or pixel and contains the intensity value. When values of x, y and intensity are discrete, then the image is said to be a digital image. Mathematically, a digital image is a matrix representation of a two-dimensional image using a finite number of points cell elements, usually referred to as pixels (picture elements, or pels). Each pixel is represented by numerical values: for grayscale images, a single value representing the intensity of the pixel (usually in a $[0, 255]$ range) is enough; for color images, three values (representing the amount of red (R), green (G), and blue (B)) are stored. If an image has only two intensities, then the image is known as a binary image. [Figure 1.1](#) shows a color image and its red, green and blue components. The color image is a combination of these three images. [Figure 1.1e](#) shows the 8-bit grayscale image corresponding to color image shown in [Fig. 1.1a](#). [Figure 1.1](#) also shows the matrix representation of a small part of these images.

MATLAB® supports the following image types:

1. **Grayscale:** A grayscale image, having $M \times N$ pixels is represented as a matrix of double datatype of $M \times N$ size in MATLAB. Element values denote the pixel grayscale intensities in the range $[0, 1]$ with 1 = white and 0 = black.
2. **True-color RGB:** A true-color red-green-blue (RGB) image is represented as a three-dimensional $M \times N \times 3$ double matrix in MATLAB. Each pixel has

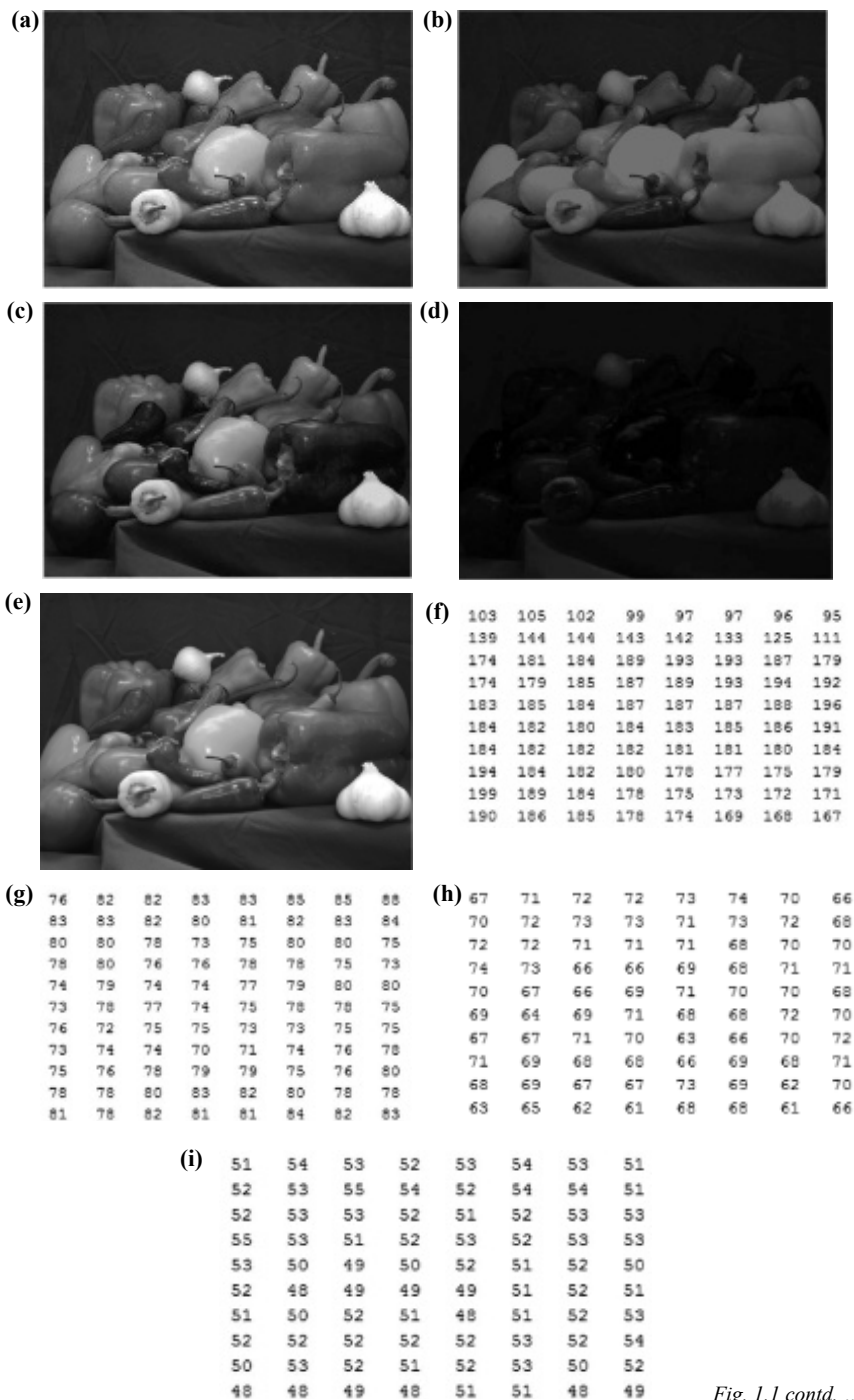


Fig. 1.1 contd. ...

red, green and blue components. The color components of a pixel (m,n) are denoted as $(m,n,1)$ = red, $(m,n,2)$ = green, $(m,n,3)$ = blue.

3. **Indexed:** In MATLAB, Indexed (paletted) images are represented with an index matrix of size $M \times N$ and a colormap matrix of size $K \times 3$. The colormap matrix holds all colors used in the image and the index matrix represents the pixels by referring to colors in the colormap.
4. **Binary:** In MATLAB, a binary image having two values, 1 (white) or 0 (black), is represented by an $M \times N$ logical matrix.
5. **uint8:** In MATLAB, this type uses less memory and some operations compute faster than with double types.

In image processing operations, most of the operations are performed in grayscale images. For color image processing applications, a color image can be decomposed into Red, Green and Blue components and each component is processed independently as a grayscale image. For processing, an indexed image is converted to grayscale image or RGB color image for most operations.

MATLAB commands `imread` and `imwrite` are used for reading and writing image files as:

`I = imread(filename);`

`imwrite(I, filename)` writes image data `I` to the file specified by `filename`, inferring the file format from the extension. The bit depth of the output image depends on the data type of `I` and the file format. For most formats:

- If `I` is of data type `uint8`, then `imwrite` outputs 8-bit values.
- If `I` is of data type `uint16` and the output file format supports 16-bit data (JPEG, PNG and TIFF), then `imwrite` outputs 16-bit values. If the output file format does not support 16-bit data, then `imwrite` returns an error.
- If `I` is a grayscale or RGB color image of double or single data type, then `imwrite` assumes that the dynamic range is $[0,1]$ and automatically scales the data by 255 before writing it to the file as 8-bit values. If the data in `I` is single, convert `I` to double before writing to a GIF or TIFF file.
- If `I` is of logical data type, then `imwrite` assumes that the data is a binary image and writes it to the file with a bit depth of 1, if the format allows it. BMP, PNG, or TIFF formats accept binary images as input arrays.

...Fig. 1.1 contd.

Fig. 1.1. (a) A color image; (b) Red Component of color image (a); (c) Green Component of color image (a); (d) Blue Component of color image (a); (e) Color image (a) converted into 8-bit grayscale image; (f) Matrix representation of upper left corner of image (b); (g) Matrix representation of upper left corner of image (c); (h) Matrix representation of upper left corner of image (d); (i) Matrix representation of upper left corner of image (e).

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(For color images of Fig. 1.1(a), (b), (c), (d) see Color Figures Section at the end of the book)

In literature, the following three levels of image processing operations are defined:

- **Low-Level image processing:** Primitive operations on images (e.g., contrast enhancement, noise reduction, etc.) are under this category, where both the input and the output are images.
- **Mid-Level image processing:** In this category, operations involving extraction of attributes (e.g., edges, contours, regions, etc.), from images are included.
- **High-Level image processing:** This category involves complex image processing operations related to analysis and interpretation of the contents of a scene for some decision making.

Image processing involves many disciplines, mainly computer science, mathematics, psychology and physics. Other areas, such as artificial intelligence, pattern recognition, machine learning, and human vision, are also involved in image processing.

1.2 Typical Image Processing Operations

Image processing involves a number of techniques and algorithms. The most representative image processing operations are:

- **Binarization:** Many image processing tasks can be performed by converting a color image or a grayscale image into binary in order to simplify and speed up processing. Conversion of a color or grayscale image to a binary image having only two levels of gray (black and white) is known as binarization.
- **Smoothing:** A technique that is used to blur or smoothen the details of objects in an image.
- **Sharpening:** Image processing techniques, by which the edges and fine details of objects in an image are enhanced for human viewing, are known as sharpening techniques.
- **Noise Removal and De-blurring:** Before processing, the amount of noise in images is reduced using noise removal filters. Image removal technique can sometimes be used, depending on the type of noise or blur in the image.
- **Edge Extraction:** To find various objects before analyzing image contents, edge extraction is performed.
- **Segmentation:** The process of dividing an image into various parts is known as segmentation. For object recognition and classification segmentation is a pre-processing step.

1.3 History of Digital Image Processing

Earlier digital image processing was mainly used in the newspaper industry to make images look better or for simply converting black and white images into color images. In the 1920s, digital images were transmitted electronically between London and New York. Initial Bartlane cable picture systems were capable of coding

an image using five gray levels; this was later enhanced to 15 gray levels in 1929. Actual digital image processing started after the invention of digital computers and related technologies, including image storage, display and transmission. In the 1960s, powerful computers gave birth to meaningful digital image processing. Satellite Images of the moon, taken by Ranger 7 U.S. spacecraft, were processed at the Jet Propulsion laboratory at California. At the same time, use of digital image processing began in various activities relating to astronomy, medical image processing, remote sensing, etc. From 1960 onwards, the use of digital image processing techniques has grown tremendously. These techniques are now used in almost every part of our life. They have applications in the fields of defence, astronomy, medicine, law, etc.

Some common examples of digital image processing are fingerprint recognition, processing of satellite images, weather prediction, character recognition, face recognition, product inspection and assembly.

1.4 Human Visual System

The human visual system consists of two parts: eye and brain. The human eye acts as a receptor of images by capturing light and converting it into signals. These signals are then transmitted to the brain for further analysis. Eyes and brain work in combination to form a picture.

The human eye is analogous to a camera. The structure of human eye is shown in Fig. 1.2:

Various parts of the human eye are identified:

- Primary Lens: Cornea and aqueous humour, used to focus incoming light signal.
- Iris: The iris dynamically controls the amount of light entering the eye, so that the eye can function in various lighting conditions, from dim light to very bright light. The portion of the lens not covered by the iris is called the pupil.
- Zonula: This is a muscle that controls the shape and positioning (forward and backward) of the eye's lens.

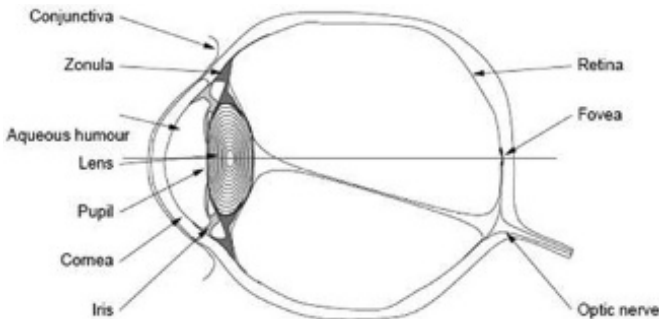


Fig. 1.2. Human eye.

- Retina: provides a photo-sensitive screen at the back of the eye; it converts the light hitting the retina into nerve signals.
- Fovea: A small central region of the retina that contains a large number of photo-sensitive cells and provides very good resolution.
- Optic nerve: These nerves carry the signals generated by the retina to the brain.

Human vision has a “blind spot” in the area of the retina where the optic nerves are attached. This blind spot does not have any photosensitive cells.

Light sensitive cells in the brain are of two types: rods and cones. There are about 120 million rod cells and 6 million cone cells. Rod cells provide visual capability at very low light levels and are very sensitive. Cone cells provide our daytime visual facilities and perform best at normal light levels.

Rods and cones are not uniformly distributed across the retina. The cones are concentrated in the center, while the rods are away from the center (Fig. 1.3). A yellow spot (macula), of size 2.5 to 3 mm in diameter, is found in the middle of the retina. Cones are very tightly packed together and the blood vessels within the fovea and other cells are pulled aside in order to expose them directly to the light.

In dim light, such as moonlight, rods in our eyes are activated and the fovea effectively acts as a second blindspot. To see small objects at night, one must shift the vision slightly to one side, around 4 to 12 degrees, so that the light falls on some rods.

Although there are about 120 million rods and 6 million cone cells in the retina, there are less than a million optic nerve fibres which connect them to the brain. This means that there cannot be a single one-to-one connection between the photoreceptors and the nerve fibres. The number of receptors connecting to each fibre is location dependent.

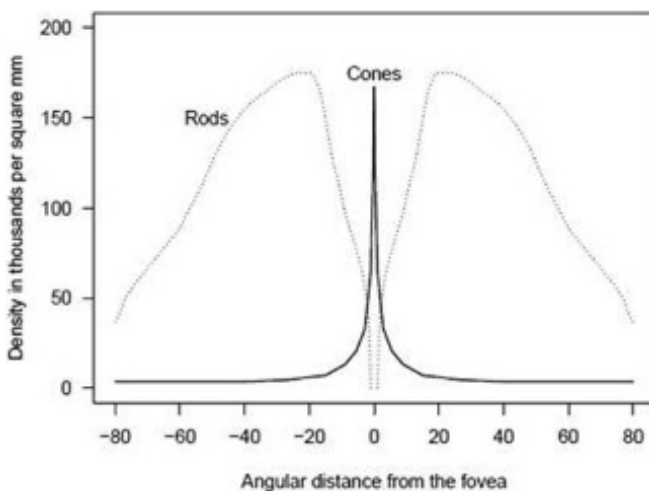


Fig. 1.3. Rods and Cones cells distribution in the retina.

In the fovea, cones have a one-to-one interaction, while in the outer part of the retina, about 600 rods are connected to each nerve fibre.

1.5 Classification of Digital Images

With regard to the manner in which they are stored, digital images can be classified into two categories: (1) Raster or Bitmap image, (2) Vector image.

A bitmap or raster image is a rectangular array of sampled values or pixels. These images have a fixed number of pixels. In the zooming of a raster image, mathematical interpolation is applied. The quality of a zoomed image degrades after a particular value of zooming factor, as shown in Fig. 1.4.

The resolution of a bitmap image is determined by the sensing device. BMP, GIF, PNG, TIFF and JPEG common image formats are bitmap or raster image formats.

On the other hand, vector images are stored in the form of mathematical lines and curves. Information like length, color, thickness, etc., is stored in the form of a vector. These images can be displayed in any size, any resolution and on any output

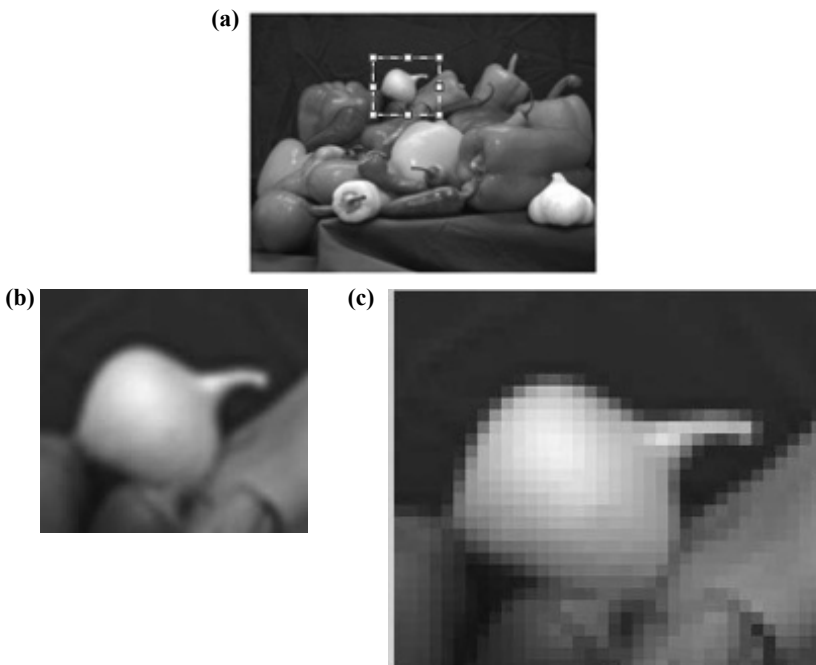


Fig. 1.4. Zooming of a Bitmap image (a) Original image (b) Part of image zoomed 4 times (c) Same part of the image zoomed 8 times.

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(For color image of Fig. 1.4(a), see Color Figures Section at the end of the book)

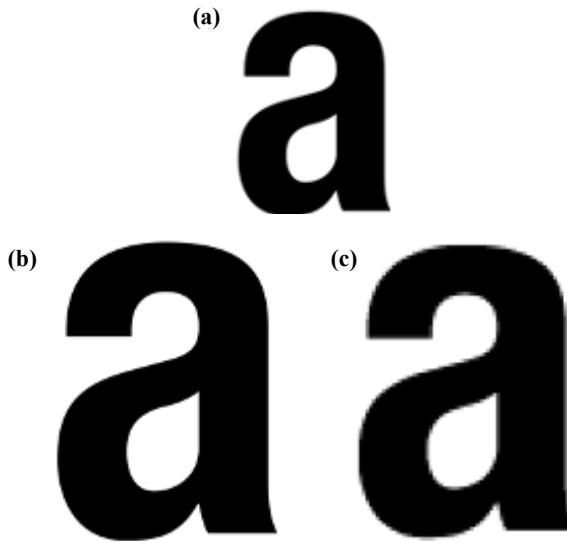


Fig. 1.5. (a) An image (b) Image in (a) zoomed in vector format (c) Image in (a) zoomed in Raster format.

device. Vector images are suitable in illustrations, line art, font, etc. The difference in zooming of a vector image and raster image can be observed in [Fig. 1.5](#). The degradation in the quality due to zooming is clearly visible on the boundaries of the character stored in raster format.

1.6 Digital Image File Types

There are a number of digital image file types available these days. The most commonly used image file types are: JPEG, GIF, TIFF, PNG and BMP. Image file types are based on the compression technique used for reducing the size of the image file. Images in various file types may differ in color, if color has been used. An image in its simplest form may contain only two intensities, i.e., black and white, and needs only 1 bit to represent intensity at each pixel.

- **TIFF** (Tagged Image File Format): This format is a very flexible and may be based on a lossy or lossless compression technique. The details of the compression technique are stored in the image itself. Generally, TIFF files use a lossless image storage format and, hence, are quite large in size.
- **Portable Network Graphics** (PNG): This format is a lossless storage format and uses patterns in the image to compress the image. The compression in PNG files is exactly reversible, meaning the uncompressed image is identical to the original image.
- **Graphical Interchange Format** (GIF): This format creates a table of upto 256 colors from 16 million colors. If the image to be compressed has less

than 256 colors, then the GIF image has exactly the same color. But if the number of colors is greater than 256, then the GIF approximates the colors in the image using a table of the 256 colors that are available.

- **Joint Picture Experts Group** (JPG or JPEG): This format is an optimized format for photographs and continuous tone images that contain a large number of colors. JPEG files can achieve high compression ratios while maintaining clear image quality.
- **RAW:** This is a lossless image format available on some digital cameras. These files are very small but the format is manufacturer dependent, therefore, the manufacturer's software is required in order to view the images.
- **Bitmapped Image** (BMP) is an uncompressed proprietary format invented by Microsoft.

Some other common image file formats are PSD, PSP, etc., which are proprietary formats used by graphics programs. Photoshop's files have the PSD extension, while Paintshop Pro files use the PSP extension.

MATLAB supports a number of image file formats, e.g., Windows Bitmap, Windows Cursor resources, Flexible Image Transport, Graphics Interchange Format, Windows Icon resources, JPEG, Portable Bitmap, Windows Paintbrush, Tagged Image File Format.

The details can be obtained using MATLAB function `imformats`. The `imformats` function can be used if the file format exists in the registry as:

`formatStruct = imformats ('bmp')`, then the output will look like:

`format Struct = struct with fields:`

```
ext: {'bmp'}
isa: @isbmp
info: @imbmpinfo
read: @readbmp
write: @writebmp
alpha: 0
description: 'Windows Bitmap'
```

1.7 Components of an Image Processing System

A complete digital image processing system comprises of various elements, such as image acquisition, storage, image processing, displays, etc. (Fig. 1.6).

Sensing devices are used to capture the image. The sensing device senses the energy radiated by the object and converts it into digital form. For example, a digital camera senses the light intensity and converts into the digital image form. Image processing elements are used to perform various operations on a digital image. It requires a combination of hardware and software. Storage is a very important part of an image processing system. The size of an image or video file is very large. For instance, an 8-bit image having 1024 x 1024 pixels requires 1 megabyte of storage space. Therefore, mass storage devices are required in image processing systems.

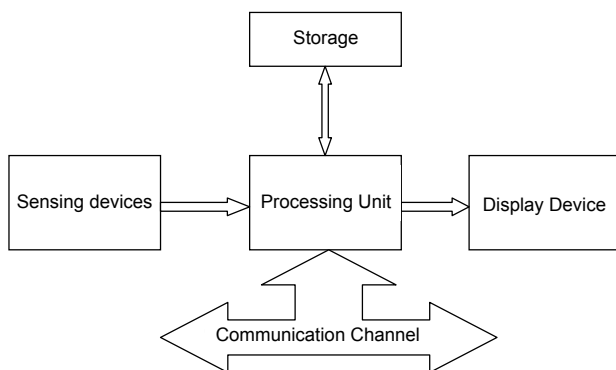


Fig. 1.6. A digital image processing system.

Display devices are required to display the images. These can be a computer monitor, mobile screen, projector for display or hardcopy devices, such as printers. A communication channel is also essential for sending and receiving images.

1.8 Applications of Digital Image Processing

Digital image processing techniques are now used in a number of applications; some common applications are given below.

In medicine: Several medical tools use image processing for various purposes, such as image enhancement, image compression, object recognition, etc. X-radiation (X-rays), computed tomography scan (CT scan), positron-emission tomography (PET), Single-photon emission computed tomography (SPECT), nuclear magnetic resonance (NMR) spectroscopy and Ultra-Sonography are some popular pieces of medical equipment based on image processing.

In agriculture: Image processing plays a vital role in the field of agriculture. Various paramount tasks such as weed detection, food grading, harvest control and fruit picking are done automatically with the help of image processing. Irrigated land mapping, determination of vegetation indices, canopy measurement, etc., are possible with good accuracy through the use of imaging techniques in various spectrums, such as hyper spectral imaging, infrared, etc.

In weather forecasting: Image processing also plays a crucial role in weather forecasting, such as prediction of rainfall, hailstorms, flooding. Meteorological radars are widely used to detect rainfall clouds and, based on this information, systems predict immediate rainfall intensity.

In photography and film: Retouched and spliced photos are extensively used in newspapers and magazines for the purpose of picture quality enhancement. In movies, many complex scenes are created with image and video editing tools which are based on image and video processing operations. Image processing-based

methods are used to predict the success of upcoming movies. For a global media and entertainment company, Latent View extracted over 6000 movie posters from IMDB along with their metadata (genre, cast, production, ratings, etc.), in order to predict the movies' success using image analytics. The colour schemes and objects in the movie posters were analyzed using Machine Learning (ML) algorithms and image processing techniques.

In entertainment and social media: Face detection and recognition are widely used in social networking sites where, as soon as a user uploads the photograph, the system automatically identifies and gives suggestion to tag the person by name.

In security: Biometric verification systems provide a high level of authenticity and confidentiality. Biometric verification techniques are used for recognition of humans based on their behaviours or characteristics. To create alerts for particularly undesirable behaviour, video surveillance systems are being employed in order to analyze peoples' movements and activities. Several banks and other departments are using these image processing-based video surveillance systems in order to detect undesired activities.

In banking and finance: The use of image processing-based techniques is rapidly increasing in the field of financial services and banking. 'Remote deposit capture' is a banking facility that allows customers to deposit checks electronically using mobile devices or scanners. The data from the check image is extracted and used in place of a physical check. Face detection is also being used in the bank customer authentication process. Some banks use 'facial-biometric' to protect sensitive information. Signature verification and recognition also plays a significant role in authenticating the signature of the customers. However, a robust system used to verify handwritten signatures is still in need of development. This process has many challenges because handwritten signatures are imprecise in nature, as their corners are not always sharp, lines are not perfectly straight, and curves are not necessarily smooth.

In marketing and advertisement: Some companies are using image-sharing through social media in order to track the influence of the company's latest products/ advertisement. The tourist department uses images to advertise tourist destinations.

In defence: Image processing, along with artificial intelligence, is contributing to defence based on two fundamental needs of the military: one is autonomous operation and the other is the use of outputs from a diverse range of sophisticated sensors for predicting danger/threats. In the Iran-Iraq war, remote sensing technologies were employed for the reconnaissance of enemy territory. Satellite images are analyzed in order to detect, locate and destroy weapons and defence systems used by enemy forces.

In industrial automation: An unprecedented use of image processing has been seen in industrial automation. The 'Automation of assembly lines' system detects the position and orientation of the components. Bolting robots are used to detect

the moving bolts. Automation of inspection of surface imperfection is possible due to image processing. The main objectives are to determine object quality and detect any abnormality in the products. Many industries also use classification of products by shape automation.

In forensics: Tampered documents are widely used in criminal and civil cases, such as contested wills, financial paper work and professional business documentation. Documents like passports and driving licenses are frequently tampered with in order to be used illegally as identification proof. Forensic departments have to identify the authenticity of such suspicious documents. Identifying document forgery becomes increasingly challenging due to the availability of advanced document-editing tools. The forger uses the latest technology to perfect his art. Computer scan documents are copied from one document to another to make them genuine. Forgery is not only confined to documents, it is also gaining popularity in images. Imagery has a remarkable role in various areas, such as forensic investigation, criminal investigation, surveillance systems, intelligence systems, sports, legal services, medical imaging, insurance claims and journalism. Almost a decade ago, Iran was accused of doctoring an image from its missile tests; the image was released on the official website, Iran's Revolutionary Guard, which claimed that four missiles were heading skyward simultaneously. Almost all the newspaper and news magazine published this photo including, The Los Angeles Times, The Chicago Tribune and BBC News. Later on, it was revealed that only three missiles were launched successfully, one missile failed. The image was doctored in order to exaggerate Iran's military capabilities.

In underwater image restoration and enhancement: Underwater images are often not clear. These images have various problems, such as noise, low contrast, blurring, non-uniform lighting, etc. In order to restore visual clarity, image enhancement techniques are utilized.

Summary

- Digital images are very important in today's digital life.
- An image may be a grayscale image or a color image. A grayscale image, having only two intensities, is termed as a binary image.
- Digital images can be classified as Raster image or Vector image, with regard to the manner in which the image is stored.
- Some common image file types which are used in our daily life are JPEG, GIF, TIFF, PNG and BMP.
- The main components of an image processing system are (1) Sensing device, (2) Image processing elements, (3) Storage device and (4) Display device.

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